

# How We Reconstructed the Neighborhood Destroyed by the Tulsa Race Massacre

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## 1 INTRODUCTION

As the 100th anniversary of the Tulsa Race Massacre approached, we set out to uncover the full scale of this obliteration — a story not lost to history but deliberately buried in it. We wanted to help readers understand what was lost when an angry white mob destroyed Greenwood — a thriving Black neighborhood in Tulsa, Oklahoma — by guiding them through a spatial narrative of success and entrepreneurship accomplished by Greenwood’s Black citizens just six decades out of enslavement.

“WhattheTulsaRaceMassacreDestroyed” is our immersive reconstruction of historic Greenwood and the devastation that it suffered. To bring this pre-1921 neighborhood back to life, we spent months analyzing archival maps, city directories, census data, newspaper clippings and survivor accounts as well as reading books, interviewing descendants of survivors and reporting on the ground. The research culminated in a detailed 3-D model of pre-massacre Greenwood, created using a series of computerized and manual steps that transformed historical material into digital data.

## 2 OUR PROCESS

Sanborn fire insurance maps from 1915 and 1920 were among the first resources we looked at. They contained building footprints, heights, materials, transportation infrastructure and much more. Some details were lost to age and mold, so AnjaliSinghvi, who led the project with YuliyaParshina-Kottas, retouched the digitized pages and stitched them together into a single image. To create a basis for the 3-D model, TimWallace and JugalPatel cleaned and painstakingly georeferenced maps from the 1915 atlas for the areas outside Greenwood and downtown Tulsa and combined the collection with the 1920 map of Greenwood.

To turn the flat maps into 3D geometry, Lingdong Huang trained a neural network to identify building footprints and convert them into extruded polygons. In order for the algorithm to learn, a series of mock maps were procedurally generated in the same style as



Figure 1: Training the neural network by The New York Times. Map source: 1915 Sanborn fire insurance maps via Library of Congress.



Figure 2: Final polygon output, based on the 1915 Sanborn map by The New York Times. Map source: 1915 Sanborn fire insurance maps via Library of Congress.

the original Sanborns. Corresponding black and white images were used to help the algorithm distinguish between building footprints and other map details. (Figure 1) Once the network had been trained, it correctly translated the Sanborn maps into crisp geometry that we used to create the final 3-D map. (Figure 2) The algorithm had difficulty identifying the building height annotations in the maps, so Mr. Huang created a Python tool to manually enter the data for each building.

To show what was lost, we knew we had to show how much Greenwood had thrived. That required specificity. No other stretch

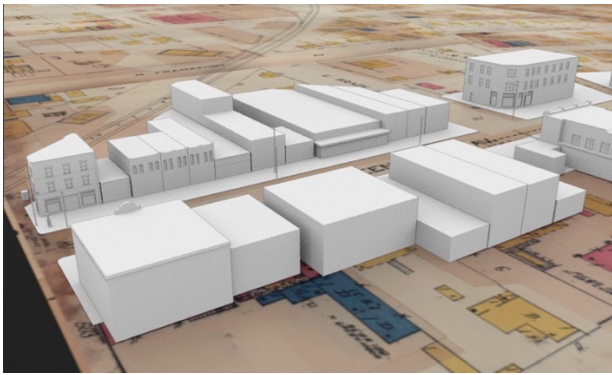
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**Figure 3: Manually constructed model of the 100 block of Greenwood Avenue by The New York Times. Map source: 1915 Sanborn fire insurance maps via Library of Congress.**

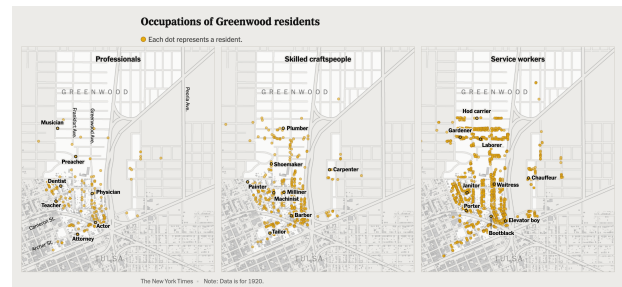
of business tells the story of Black entrepreneurship better than the 100 block of Greenwood Avenue. So we highlighted this marquee block and its bustling retail scene.

Finding visual references of pre-massacre Greenwood required substantial detective work because many of the photographs taken before 1921 had been destroyed in the massacre. MikaGröndahl and GuilbertGates confirmed two pre-massacre photographs of Greenwood Avenue, offering an incomplete view of the street, as well as several closeups of buildings on the block. Lacking sufficient references, they turned to images taken during and immediately after the massacre. Mr. Gröndahl and Mr. Gates spent hours sifting through photos of blocks that had been completely destroyed. They looked for details that might help them understand what had stood there. Barber poles, wall fragments and floor joists provided hints about what the buildings had been before they were burned.

Mr. Gröndahl, a veteran 3-D artist and a trained architect, manually crafted detailed building models in Maya based on the archival photographs. (Figure 3) Using camera projection mapping, he aligned photo references with our 3-D environment, calibrating the focal length, type of lens, location of camera, filmback size and building dimensions, to produce the most accurate model possible. When no visual references were available, we left the buildings blank so we wouldn't be representing anything that we hadn't verified.

We set out to not only model structures, but also replicate a feeling of moving through space. As a reader, you can make your way down the street and pass by the four hotels, the theater, the cabaret, the seven barbers, the nine restaurants, and the offices of eight doctors and a half-dozen real estate agents, dentists and lawyers. We pored over old photographs, clippings from Black-owned newspapers and other archival material to verify the addresses of these shops and match them to their owners. In the interactive, you can also meet the townspeople, like Loula and John Williams, who owned a confectionery and the 750-seat Williams Dreamland Theater. You learn of Mary E. Jones Parrish's typing school and Mabel B. Little's salon.

For a deeper exploration of the lives led by Greenwood residents in the neighborhood's 35 blocks, we created maps of their



**Figure 4: Maps of occupations of Greenwood residents by The New York Times. Data source: 1920 Tulsa census via Ancestry.**

occupations after analyzing handwritten 1920 U.S. census records, digitized by Ancestry.com. (Figure 4) We cleaned the data to include the correct street names from 1921 Greenwood and manually geocoded thousands of addresses. This allowed us to visualize the occupations by job type.

A data set of Black businesses and residents was created by running the scanned pages of the 1921TulsaCityDirectory through an Optical Character Recognition process and searching the results for a "(c)" at the end of an entry (used by the directory to identify businesses and residents of color). Additional information about businesses and proprietors on our featured block was gathered through manual review of archival newspapers and documents.

JeremyWhite wrote a custom After Effects script to convert 3D tracking information for text labels into 2D screen-space and synchronize rendered content to scroll events. This allowed us to tailor video resolutions based on display devices while maintaining the sharpness and spatial placement of text labels in the scene.

We released all of the datafilesassociatedwiththetory to the public so that others could build on the work that we published and further explore what was once the most prosperous Black community in the United States.